Panic Attack Dataset Analysis

Detailed Project Report

Tools: Snowflake SQL, Power BI Desktop
Data connection: CSV loaded to Snowflake, then imported into Power BI using Import Mode
Dataset: `panic_attack_dataset.csv`
Records: 1,200
Columns: 21
Project type: Descriptive data analysis and interactive business intelligence reporting

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1. Executive Summary

This project explores 1,200 sample records to understand recorded panic-attack patterns across demographics, trigger categories, duration groups, sleep-hour categories, medical-history categories, and medication/therapy combinations. Snowflake SQL was used for data profiling, validation, aggregation, and comparison. Power BI was used in Import Mode to create a four-page interactive report.

The dataset has an average recorded duration of 24.39 minutes and an average panic score of 5.57. The 46–55 age group has the largest record count (272), while the Medium duration group (15–24 minutes) has the largest duration-category count (323). These are descriptive observations from this sample dataset and do not establish causes or medical conclusions.

2. Project Background and Objective

The aim was to practise an end-to-end analytics workflow: understand the dataset, validate its quality, use SQL to answer focused questions, and communicate results through a Power BI dashboard.

Objectives

- Profile the dataset and check record quality.
- Calculate descriptive statistics for duration, panic score, and attack frequency.
- Compare duration and panic score across trigger, gender, age, sleep, and treatment-code groups.
- Review medical-history category counts.
- Build an interactive Power BI report with an overview and focused analysis pages.
- Communicate findings with clear limits and data caveats.

3. Data Source and Architecture

Workflow

1. Source file: `panic_attack_dataset.csv`.
2. Load the CSV into Snowflake.
3. Query and analyse the table in Snowflake SQL Worksheet.
4. Connect Power BI Desktop to Snowflake.
5. Select **Import** mode to load data into the Power BI model.
6. Build calculated columns, measures, charts, slicers, and findings in Power BI.

Important: In Import Mode, Power BI works with an imported copy of the data. Refresh the model after source changes if the report needs updated data.

4. Dataset Overview

The file contains 1,200 rows and 21 columns.

Area	**Fields**
Identifier	`ID`
Demographics	`Age`, `Gender`
Attack measures	`Panic_Attack_Frequency`, `Duration_Minutes`, `Panic_Score`
Trigger	`Trigger`
Physical measures and symptoms	`Heart_Rate`, `Sweating`, `Shortness_of_Breath`, `Dizziness`, `Chest_Pain`, `Trembling`
Medical/treatment	`Medical_History`, `Medication`, `Therapy`
Lifestyle	`Caffeine_Intake`, `Exercise_Frequency`, `Sleep_Hours`, `Alcohol_Consumption`, `Smoking`

The meaning, unit, and collection method for every field have not been supplied. In particular, the unit or time period represented by `Panic_Attack_Frequency` is not documented.

5. Data Quality and Validation

Checks performed or required

- Row count and unique ID count.
- Duplicate ID and duplicate-row checks.
- Null and blank-value checks.
- Category value checks.
- Minimum, maximum, average, median, and standard deviation checks.
- Consistent, non-overlapping age and duration groups.
- Cross-checking SQL results against Power BI.

Verified findings from the uploaded CSV

- Rows: 1,200.
- Columns: 21.
- Unique IDs: 1,200.
- Duplicate IDs: none observed.
- Missing values: 122 in `Medical_History`; no missing values were observed in the other 20 columns in the uploaded CSV.
- `Medical_History` populated categories: Anxiety 492, Depression 349, PTSD 237. The remaining 122 records are blank in the CSV.

Data-quality caveat: The Power BI visual displayed a `None` category with 122 records. This appears consistent with the 122 blank source values being represented as `None` in the reporting layer, but the transformation should be confirmed. Do not describe these as a confirmed medical-history category unless the source owner defines them that way. Recommended reporting label: `Not recorded` or `Missing`.

Category-boundary correction

Use these non-overlapping age groups:

- 18–25
- 26–35
- 36–45
- 46–55
- 56–64

An earlier query used 18–24 and assigned age 25 to the next group, causing a mismatch. The corrected boundaries produce counts of 210, 232, 268, 272, and 218, respectively.

Use these duration groups:

- Short: 5–14 minutes
- Medium: 15–24 minutes
- Long: 25–34 minutes
- Very Long: 35–44 minutes

Use these sleep groups:

- Below 5 Hours: less than 5
- 5–7 Hours: 5 to less than 7
- 7+ Hours: 7 or more

6. Descriptive Statistics

Metric	Average	Median	Minimum	Maximum
Panic attack frequency	4.41	4	0	9
Duration (minutes)	24.39	24	5	44
Panic score	5.57	6	1	10

Duration standard deviation (sample): approximately 11.40 minutes (SQL `STDDEV`/`STDDEV_SAMP`; the earlier SQL output was 11.39 due to display precision).

Duration range: 39 minutes.

7. Analysis Findings

7.1 Duration group distribution

Duration group	Records	Share
Short (5–14 min)	290	24.17%
Medium (15–24 min)	323	26.92%
Long (25–34 min)	303	25.25%
Very Long (35–44 min)	284	23.67%

Duration group	Records	Share
Total	1,200	100%

Finding: Medium is the largest single duration group with 323 records (26.92%). Combined, Medium and Long account for 626 records (52.17%), between 15 and 34 minutes.

7.2 Average duration by trigger

Trigger	Records	Average duration
Caffeine	202	25.04 min
Social Anxiety	197	24.90 min
PTSD	205	24.79 min
Stress	187	24.21 min
Phobia	203	24.07 min
Unknown	206	23.34 min

Finding: Caffeine has the highest average recorded duration (25.04 minutes), while Unknown has the lowest (23.34 minutes). The difference between these averages is about 1.70 minutes, so the group averages are relatively close. This comparison does not show that a trigger causes a longer duration.

7.3 Gender comparison

Gender	Records	Average duration	Average panic score
Female	549	24.86 min	5.63
Male	537	23.97 min	5.53
Non-binary	114	24.16 min	5.43

Finding: Female records have the highest average duration (24.86 minutes) and average panic score (5.63) among the gender groups. The differences are descriptive and should not be interpreted as evidence that gender causes a difference.

7.4 Age group comparison

Age group	Records	Average duration	Average panic score
18–25	210	23.15 min	5.75
26–35	232	24.22 min	5.69
36–45	268	25.41 min	5.21
46–55	272	23.50 min	5.61
56–64	218	25.64 min	5.66

Finding: The 46–55 group has the highest record count (272, 22.67%). The 56–64 group has the highest average duration (25.64 minutes), while the 18–25 group has the highest average panic score (5.75).

7.5 Sleep-hour group comparison

Sleep group	Records	Average panic score	Average duration
Below 5 Hours	217	5.29	24.81 min
5–7 Hours	505	5.65	24.15 min
7+ Hours	478	5.61	24.46 min

Finding: The 5–7 hours group has the highest average panic score (5.65). The Below 5 Hours group has the highest average duration (24.81 minutes). These comparisons do not establish that sleep duration affects panic attacks.

7.6 Medical-history category distribution

Medical-history value	Records	Share of all records
Anxiety	492	41.00%
Depression	349	29.08%
PTSD	237	19.75%

Medical-history value	Records	Share of all records
Blank / not recorded	122	10.17%
Total	1,200	100%

Finding: Anxiety is the largest populated medical-history category (492 records). However, 122 records have no recorded value. These must be shown as missing/not recorded unless the source definition confirms another meaning.

7.7 Medication and therapy combination

Medication	Therapy	Records	Average panic score	Average duration
No	Yes	362	5.63	23.92 min
No	No	338	5.40	24.62 min
Yes	No	257	5.67	25.20 min
Yes	Yes	243	5.61	23.93 min

Finding: The Medication = Yes and Therapy = No group has the highest average panic score (5.67) and average duration (25.20 minutes). The combination groups are observational only. They do not measure treatment effectiveness and should not be used to recommend or compare medical treatments.

8. Power BI Dashboard Design

The report has four pages.

Page 1: Panic Attack Overview

Purpose: Give a quick summary of the dataset. Suggested visuals:

- KPI cards: Total Records, Average Duration, Average Panic Score, Average Attack Frequency.
- Gender distribution.
- Record count by trigger.
- Record count by age group.
- Duration group distribution.
- One concise executive finding.

Page 2: Attack Duration & Trigger Analysis

Purpose: Explore the duration distribution and compare average duration by trigger. Suggested visuals:

- Duration-group distribution.
- Average duration by trigger.
- Optional slicers: Gender, Age Group, Trigger.

Page 3: Demographic Analysis

Purpose: Compare record counts and descriptive metrics by age and gender. Suggested visuals:

- Record count by age group.
- Average duration by gender.
- Average panic score by gender.
- Optional slicers: Age Group, Gender, Trigger.

Page 4: Lifestyle & Medical History Analysis

Purpose: Compare sleep groups and medical-history categories. Suggested visuals:

- Average panic score by sleep category.
- Average duration by sleep category.
- Record count by medical-history category.
- Optional slicers: Sleep Category, Medical History, Gender.

Dashboard design recommendations

- Use descriptive chart titles and clear units.
- Use bar/column charts for comparing averages across categories.
- Use donut charts mainly for simple composition, such as gender share.
- Sort age groups and duration groups in their natural order.
- Use consistent colours, fonts, spacing, and slicer positions.
- Keep the Overview page focused; place detailed comparisons on analysis pages.
- Ensure all pages use the same definitions, filters, and category boundaries.

9. Limitations

- This is a sample dataset. Its real-world source, sampling method, and representativeness are not established.

- The field definitions and units are incomplete. In particular, the time period for attack frequency and the scale definition for panic score need documentation.
- The 122 blank `Medical_History` values require clear treatment in the final model.
- The meanings of any coded values must be confirmed before assigning labels.
- The dataset supports descriptive comparisons, not causal conclusions or medical advice.
- No time-series field is listed, so trends over time cannot be analysed from the supplied columns.

10. Conclusion

The project demonstrates an end-to-end descriptive analytics workflow using Snowflake and Power BI. SQL was used to validate and summarise the dataset, compare categories, and calculate key metrics. Power BI was used to communicate the results through a four-page interactive dashboard.

The main value of the project is the ability to turn raw records into clear, traceable summaries while identifying data limitations and avoiding unsupported conclusions.

Appendix: Reproducibility

The accompanying SQL file contains Snowflake-compatible queries for:

1. Dataset overview and data quality.
2. Missing values and category distributions.
3. Summary statistics.
4. Duration groups.
5. Trigger comparison.
6. Gender comparison.
7. Age group comparison.
8. Sleep group comparison.
9. Medical-history distribution.
10. Medication and therapy comparison.

Replace the table reference `PANIC_ATTACK_DATA` with the actual Snowflake database/schema/table name if it differs.